



Group B4

Smartphone Chassis FEA

MEU44B07 – Computer Aided Design

Year 4

Mechanical &
Manufacturing

Structural and Thermal Design Optimisation of a High-Performance Smartphone Chassis Through Finite Element Analysis

Problem Statement & Objectives

In November 2024, **91% of American adults owned a smartphone** – up from just 35% in 2011. However, this rapid growth has not come without design challenges. Even some of the largest manufacturers have produced products that, over time, were deemed defective:

- Apple's "Bendgate" saw users report the **iPhone 6 and 6 Plus** for bending in their pockets under relatively low loading conditions.
- Samsung's **Galaxy Note 7** was recalled after their attempts to decrease battery size and increase power capacity led to **device explosions**.

Therefore, the goal of this project was to design a phone chassis that:

- Is **lightweight**
- Is **aesthetically attractive**
- Is **stiff and strong** enough to withstand bending and torsion loads
- Uses a realistic and **suitable material**
- Efficiently **dissipates heat** generated by the phone's CPU and battery

Material Selection

After conducting research, four candidate materials were evaluated based on structural performance, thermal conductivity, and manufacturability.

Property	Ti-6Al-4V	Al 6061-T6	AZ91D	CFRTP
Density (kg/m ³)	4430	2700	1810	1650
Young's Modulus (GPa)	113.8	68.9	44.8	24
Thermal Conductivity (W/m·K)	6.7	167	72.7	8
Specific Stiffness (GPa·cm ³ /g)	25.7	25.5	24.8	15

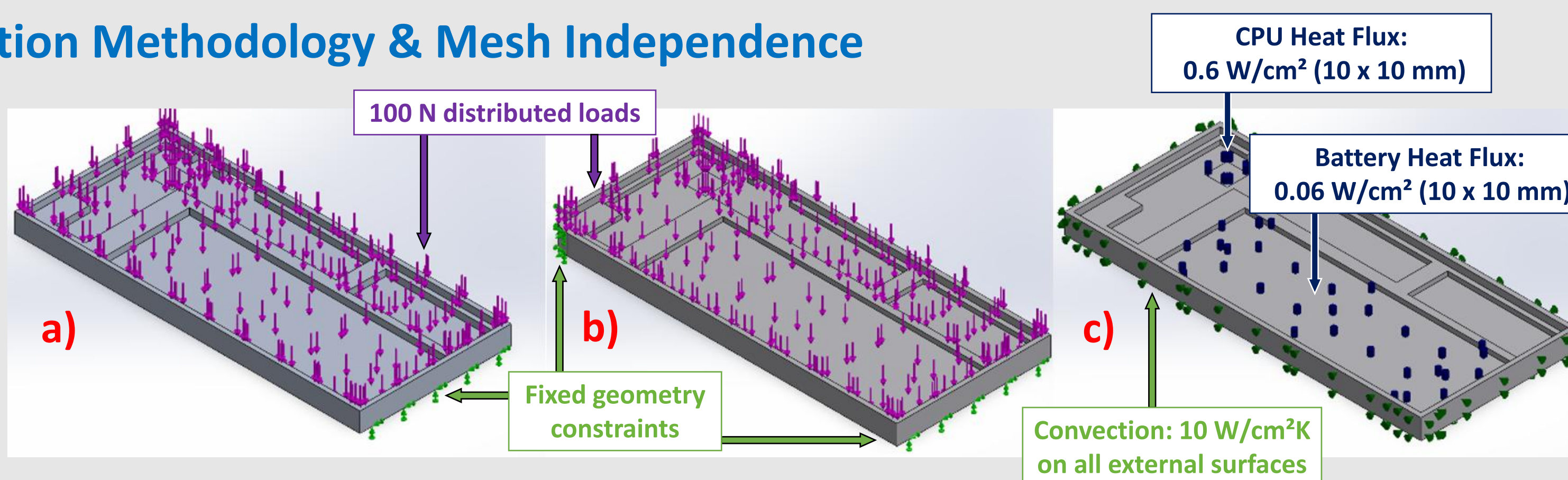
Eliminated ✗ or Shortlisted — :

✗ Titanium:	High density, poor thermal conductivity.
✗ Carbon Fibre:	Anisotropic behaviour, poor thermal conductivity.
— Magnesium (AZ91D):	Lowest density, moderate thermal performance.
Selected ✓:	
✓ Aluminium 6061-T6:	Best thermal conductivity, higher stiffness, lower strength ratio, can be anodised to improve thermal performance, better corrosion resistance.

Simulation Methodology & Mesh Independence

To begin, a 160 x 70 x 9 mm simplified chassis was used to isolate mesh effects.

- Simple bending setup
- Torsion setup
- Thermal heat flux setup

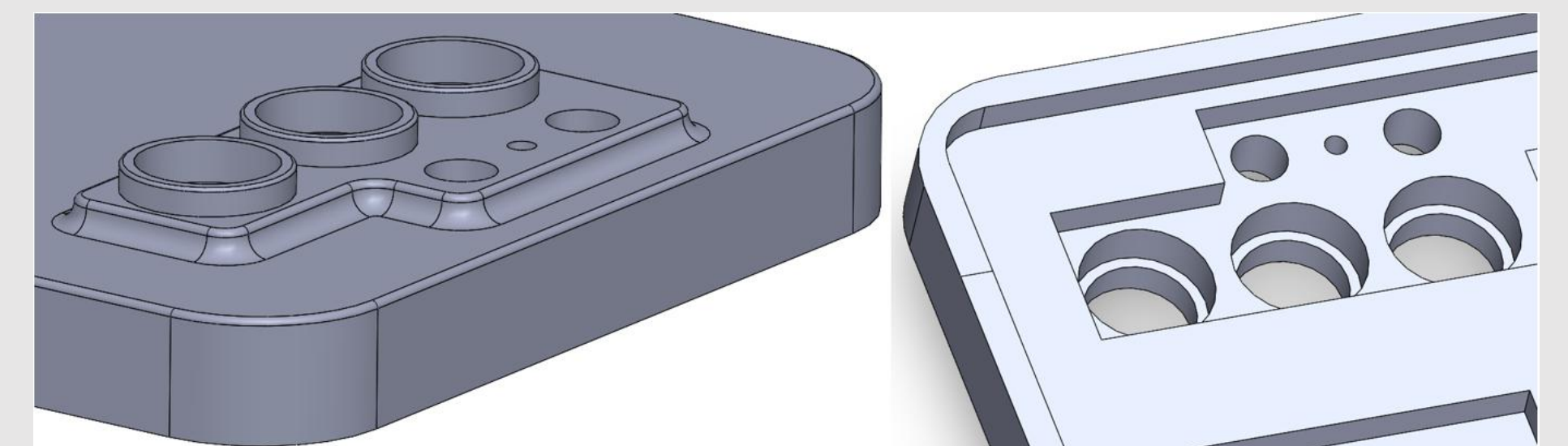


Structural Mesh Independence

Case A: Simple Bending	Case B: Torsion & Bending
Problem: - Global mesh refinement caused stress singularities at fixed edges - Max stress diverged: 14.36 MPa → 28.86 MPa (48.6% error) - Displacement and strain energy converged, stress did not	Method: - H-adaptive refinement used to automatically refine elements in high-error regions. - Target accuracy: 98% 5 refinement iterations carried out - Curvature-based meshing to capture geometry detail.
✓ Solution: - Local mesh control was applied along the chassis frame. 4 mm global + 0.8 mm local.	✓ Converged: - Displacement: 0.0645 mm (0.28%) - Strain energy: 1.499E-03 J - Strain energy norm error: 2.13% (just over 2% target)
✓ Result: - Stress converged at 14.36 MPa - Displacement: 0.085 mm 0% relative error.	✗ Did Not Converge: - Stress singularity at fixed corner (unavoidable) - Max stress differential: 125 → 2900 MPa

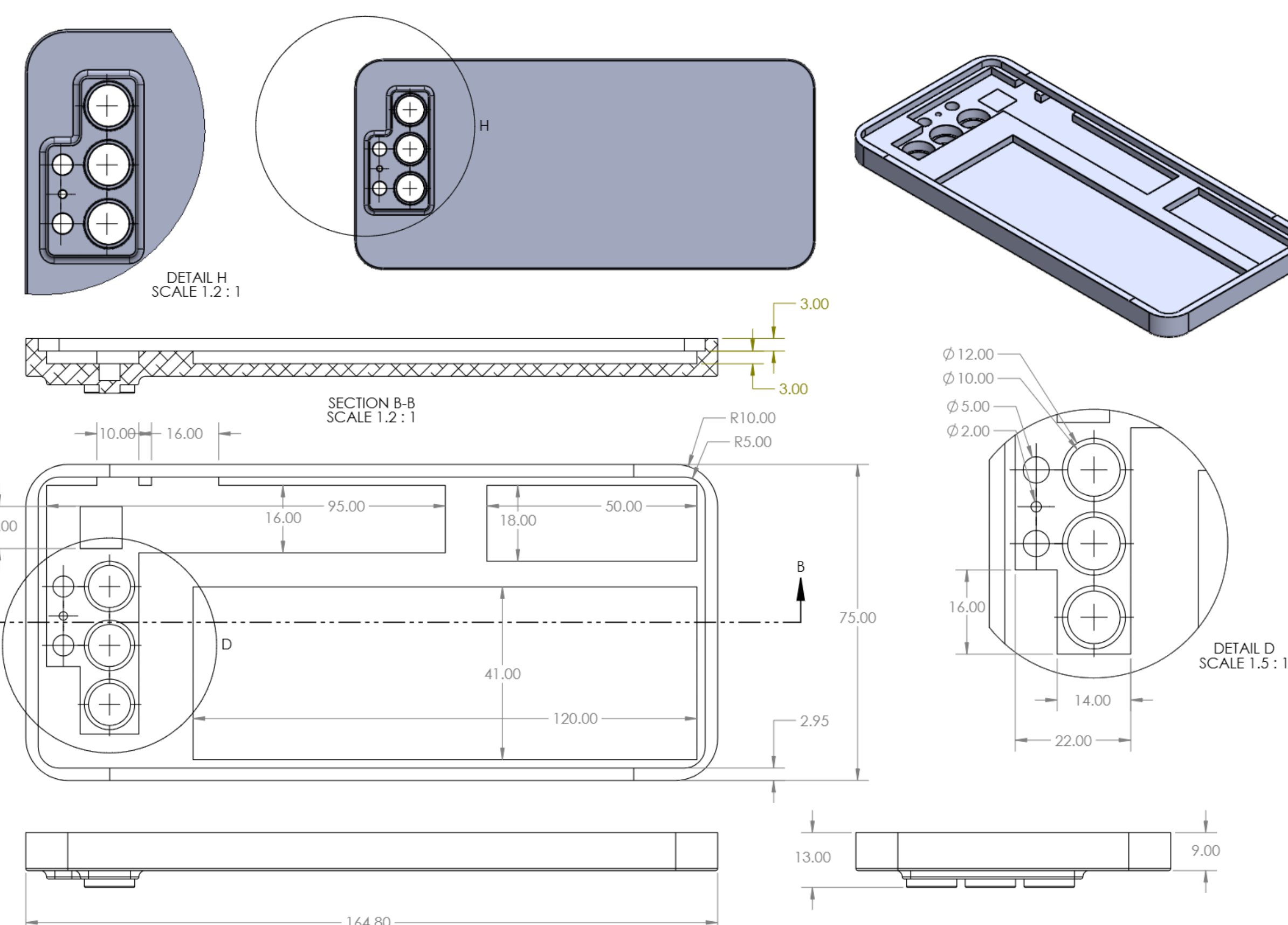
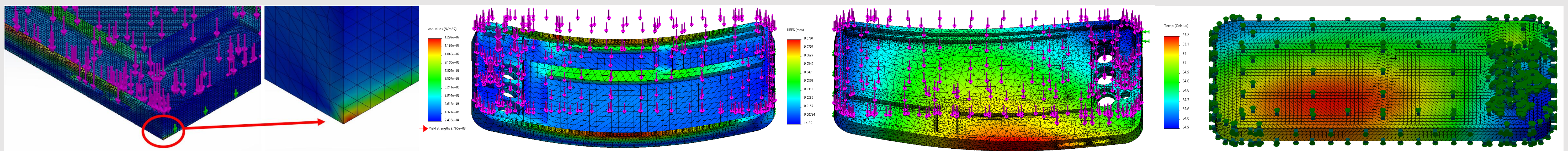
Parametric Design Optimisation

Variables:	Constraints:
- Height: 156 – 165 mm - Width: 70 – 75 mm - Depth: 8 – 9 mm - Frame Thickness: 1 – 3 mm	- Stress: $\leq 0.6 \times \sigma_{yield}$ - Displacement: ≤ 1.6 mm
Goals: - Minimise temperature - Minimise mass	
✓ Optimal Dimensions: 164.8 x 75 x 9 mm Frame: 2.95 mm	



Thermal Mesh Independence

- Refined: 4 mm → 2 mm → 1 mm - Max temp: 43.5°C (constant across all meshes) 0% relative error
✓ Adopted: 2 mm global (accuracy with low compute cost)



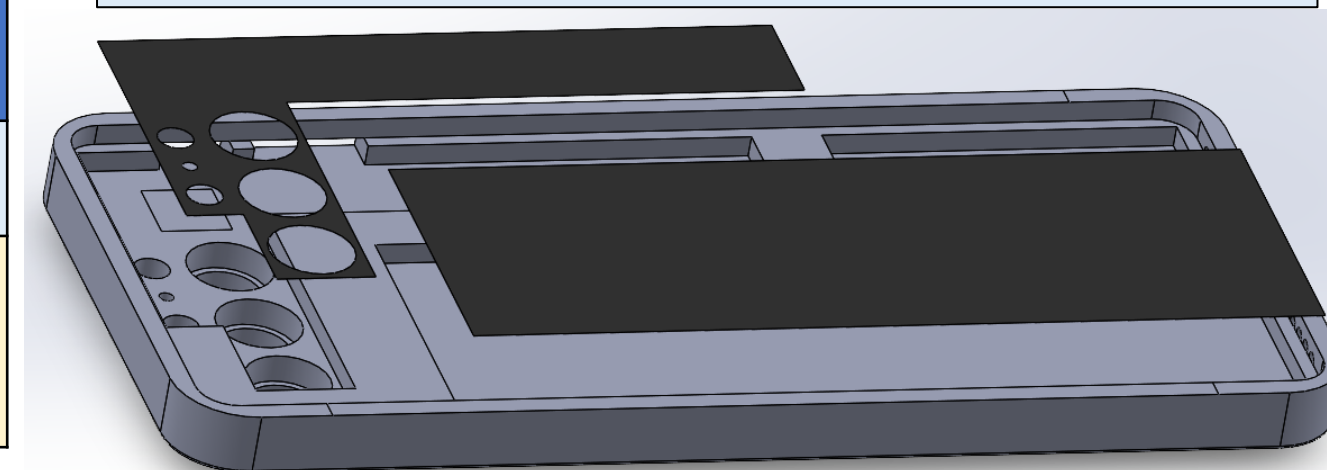
Final Design Simulation Performance

Simple Bending	Torsion	Thermal
- Max δ: 0.0802 mm - Max σ: 12.99 MPa	- Max δ: 0.0784 mm - Stiffness: 1,275 N/mm	- Min temp: 34.8°C $R_{thermal}$: 4.3 K/W

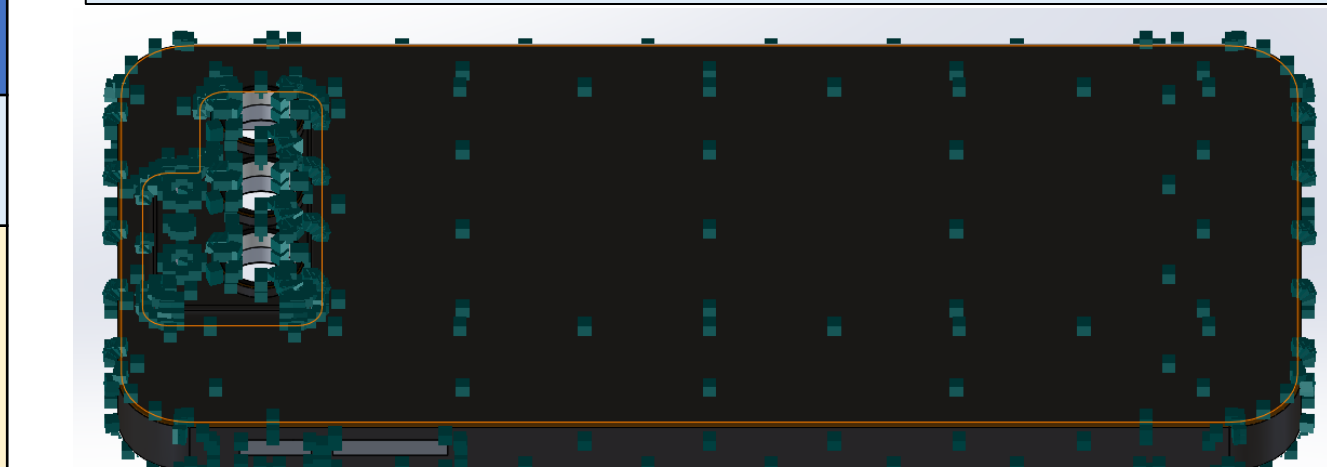
Results and Thermal Enhancements

Structural Performance	Thermal Optimisation
- Stiffness: 1,247 N/mm - Specific Stiffness: 8.7 (N/mm)/g - Mass: 143.5 g - Strength ratio: 4.7% - Buckling FOS: 400 approx.	- Baseline: 40.1°C + Graphene Layer → 39.8°C + Anodised Finish → 35.5°C + Vent perforations → 35.2°C - Max temp: 35.2°C (safe) - Total ΔT: -4.9°C

Graphene Layer (heat dispersion)



Anodised Finish (heat dissipation)



Vent Perforations (heat convection)

